Laplace

$$V_e(s) = R_1 I_1(s) + \frac{1}{Cs}[I_1(s) - I_2(s)]$$

$$\frac{1}{Cs}[I_1(s) - I_2(s)] = (Ls + R_2) I_2(s)$$

$$V_s(s) = R_2 I_2(s)$$

$$V_e(t) = R_1 i_1(t) + \frac{1}{C}\int (i_1(t) - i_2(t))dt$$

$$\frac{1}{C}\int (i_1(t) - i_2(t))dt = L\frac{di_2(t)}{dt} + R_2 i_2(t)$$

$$V_s(t) = R_2 i_2(t)$$

$$I_1(s) - I_2(s) = Cs(Ls + R_2) I_2(s)$$

$$I_1(s) = [1 + Cs(Ls + R_2)] I_2(s)$$

$$V_e(s) = R_1[1 + Cs(Ls + R_2)] I_2(s) + (Ls + R_2) I_2(s)$$

$$V_e(s) = [R_1 + R_1 Cs(Ls + R_2) + Ls + R_2] I_2(s)$$

$$V_e(s) = (R_1 LCs^2 + (L + R_1 R_2 C)s + (R_1 + R_2)) I_2(s)$$

$$\frac{V_s(s)}{V_e(s)} = \frac{R_2}{R_1 LCs^2 + (L + R_1 R_2 C)s + (R_1 + R_2)}$$

Error en estado estacionario

$$e(s) = \lim \to 0 \left[1 - \frac{R_2}{R_1 LCs^2 + (L + R_1 R_2 C)s + (R_1 + R_2)}\right]$$

$$e(s) = 1 - \frac{R_2}{R_1 + R_2}$$

$$e(s) = \frac{R_1 + \cancel{R_2} - \cancel{R_2}}{R_1 + R_2}$$

$$e(s) = \frac{R_1}{R_2 + R_1}$$

R_1	R_2		
1k	1k	Caso →	$1 - \frac{1}{1+1} = 0.5$
1k	100k	Control →	$1 - \frac{1}{1+100} = 0.99$

Estabilidad del sistema

Control = $\lambda_1 = -999989.9898896874$

$\lambda_2 = -1010.0101103126212$

$\}$ Estable con respuesta sobreamortiguada

$$\text{Caso} = \left.\begin{array}{l} \lambda_1 = -9998.9997999939978 \\ \lambda_2 = -2.00020000600022009 \end{array}\right\} \begin{array}{l}\text{Estable con respuesta} \\ \text{sobreamortiguada}\end{array}$$